

OVERVIEW

The graph class is used to represent a directed labeled graph in which each node and edge is unique from any other. Clients can add nodes and edges, as well as see a list of nodes in the graph or the children of any one node in the graph.

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Two hashsets are used to keep track of the unique edges and nodes of the graph. Each edge holds information on the source node, destination node, and its label. Each node simply contains its own data. Both the node and edge classes have “equals” methods to compare nodes and edges to one another as well as functions to create hash codes in their constructors.

INVARIANT

Each node and edge must not be null and unique from other nodes and edges.

PSOFT SPECIFICATIONS

```
/* requires: none
modifies: none
effects: Hashsets nodeCollection and EdgeCollection created
returns: none
throws: RuntimeException if a duplicate node or edge is present
*/
```

public Graph()

```
/* requires: node that is not null
modifies: hashset nodeCollection
effects: Adds node to the hashset nodeCollection
returns: none
throws: RuntimeException if a duplicate node or edge is present
*/
```

public void addNode()

```
/* requires: edge that is not null
modifies: hashset edgeCollection
effects: Adds edge to the hashset edgeCollection
returns: none
throws: RuntimeException if a duplicate node or edge is present
*/
```

public void addEdge()

```
/* requires: none  
modifies: none  
effects: none  
returns: a linked list of nodes that are present in the graph  
throws: none  
*/
```

```
public Iterator<String> listNodes()
```

```
/* requires: parent node that is not null  
modifies: none  
effects: none  
returns: a sorted linked list of children and labels of the parent node  
throws: none  
*/
```

```
public Iterator<String> getChildren(Node parent)
```